

AS6041 - Assignment 1

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1 Taylor Green Vortex Simulation

1.1 Test Case Summary

The simulation of the Taylor-Green vortex problem was performed using a 3-D, viscous, compressible Navier-Stokes solver.

Computational Domain	Cube with sides of length 2π
Grid	Equally spaced 64^3 cartesian grid
Temporal Discretisation	Runge-Kutta, 4^{th} order
Time Step	$\Delta t = 0.001$
Space Discretisation	Finite Difference Method, accuracy ranging from 2^{nd} order explicit to 6^{th} order compact
Filtering	Implicit filtering, 10^{th} order, $\alpha_f = 0.495$
Boundary Conditions	Periodic

Table 1: TGV, Test Case Summary

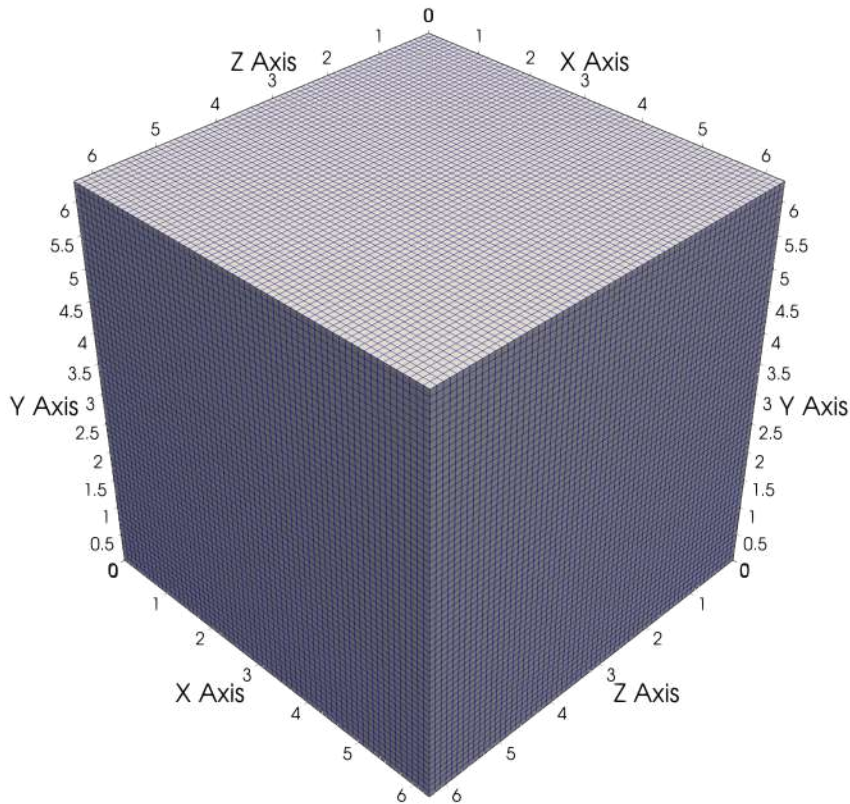


Figure 1: TGV, Computational Domain

The initial non-dimensional velocity and pressure field in the domain is

$$u = \sin(x/L) \cos(y/L) \cos(z/L) \quad (1)$$

$$v = -\cos(x/L) \sin(y/L) \cos(z/L) \quad (2)$$

$$w = 0 \quad (3)$$

$$p = \frac{1}{\gamma M^2} + \frac{1}{16} \left[\left(\cos\left(\frac{2x}{L}\right) + \cos\left(\frac{2y}{L}\right) \right) \left(\cos\left(\frac{2z}{L}\right) + 2 \right) \right] \quad (4)$$

The temperature field is taken to be uniform.

Reynolds Number	1600
Mach Number	0.1
Prandtl Number	0.71
Reference Temperature	300

Table 2: TGV, Non-dimensional flow parameters

1.2 Discussion

The variation of kinetic energy (E_k), enstrophy (ζ) and the kinetic energy dissipation rate (ϵ) with time are given below in figures 2, 3 and 4 respectively.

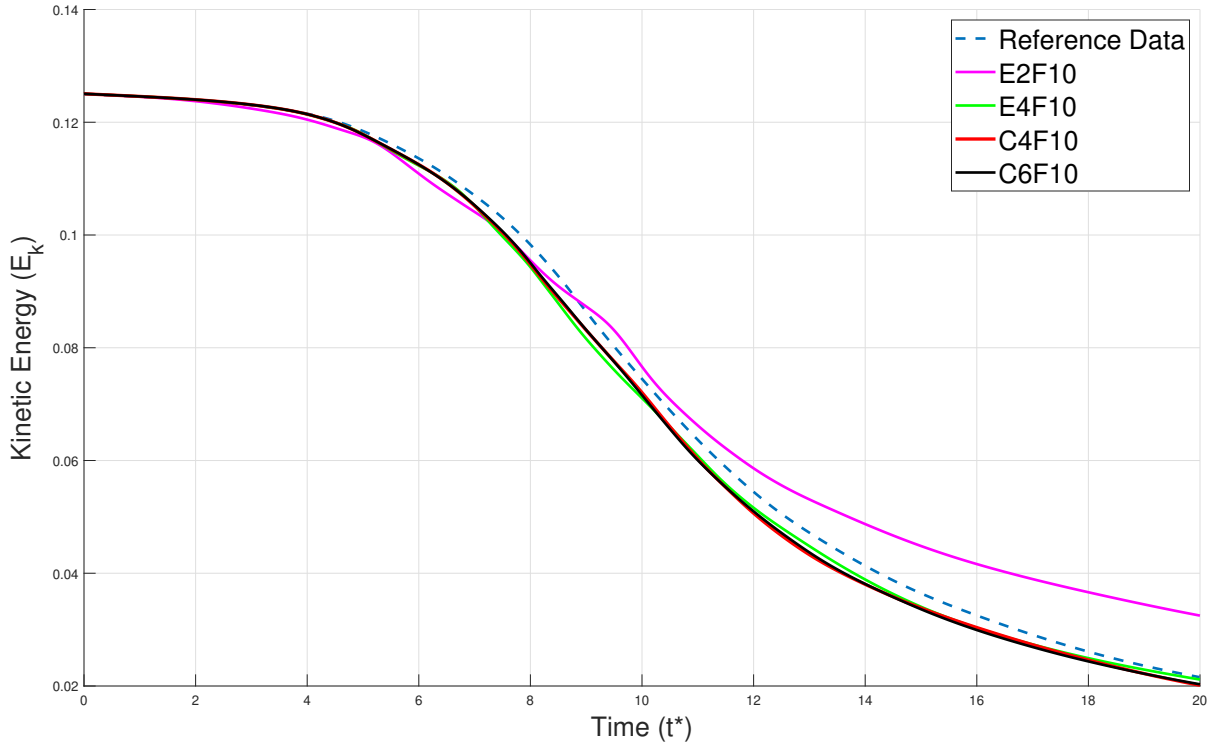


Figure 2: Kinetic Energy Variation

From figure 2 it can be seen that the kinetic energy determined by the 64^3 is in close agreement with the values of the reference data obtained on the 512^3 grid, except when the E2F10 set-up is used, due to the low accuracy of E2 scheme.

Around $t^* = 5 - 6$, the kinetic energy of obtained using 64^3 grid starts falling below the reference data, this would imply that coarser grids under-predict kinetic energy as time evolves.

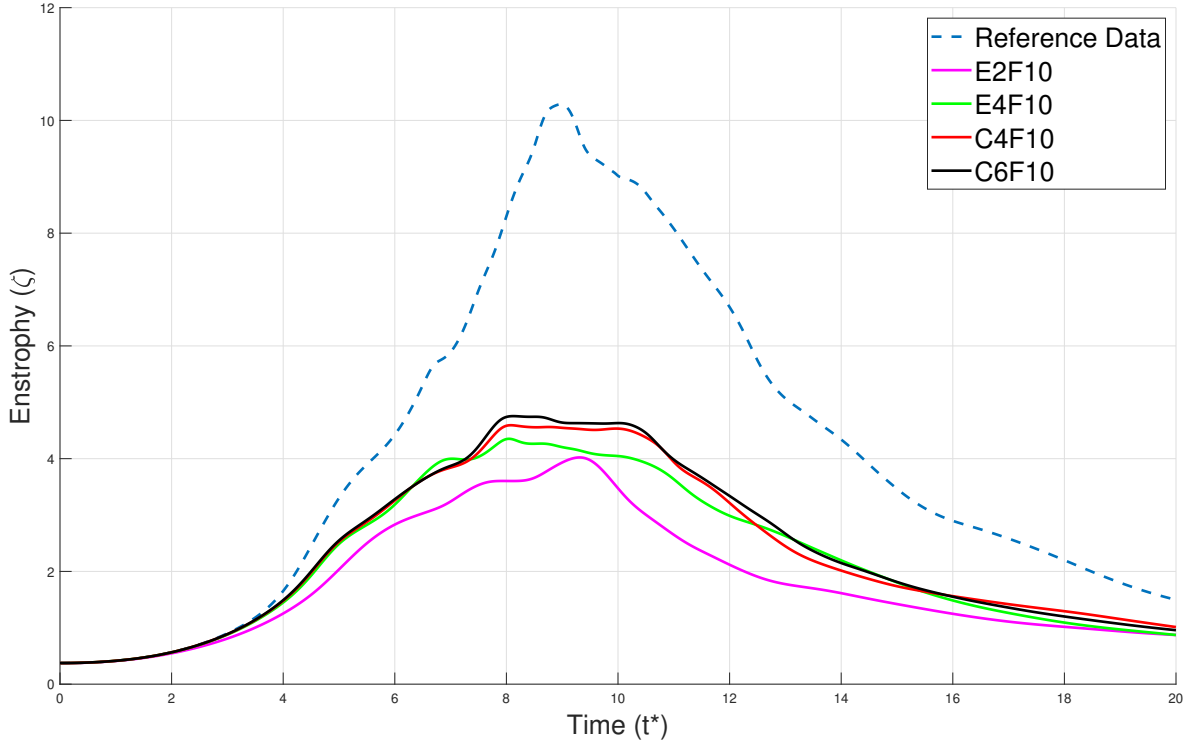


Figure 3: Enstrophy Variation

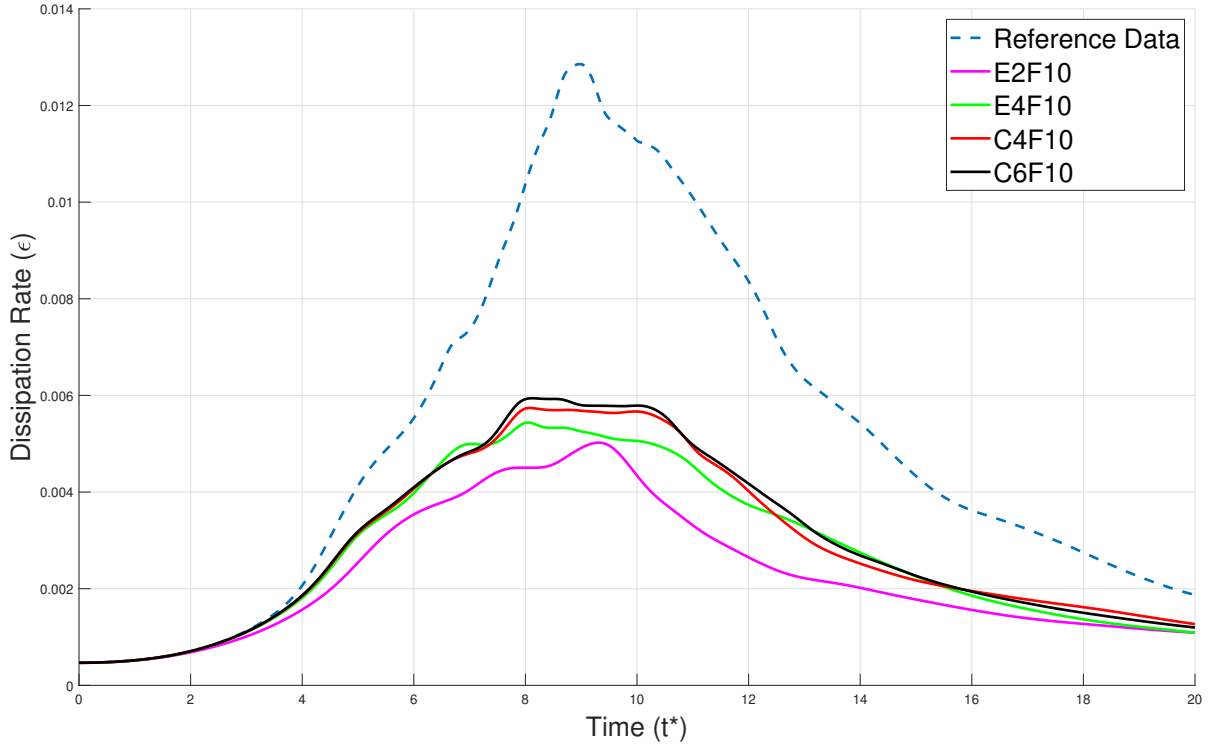
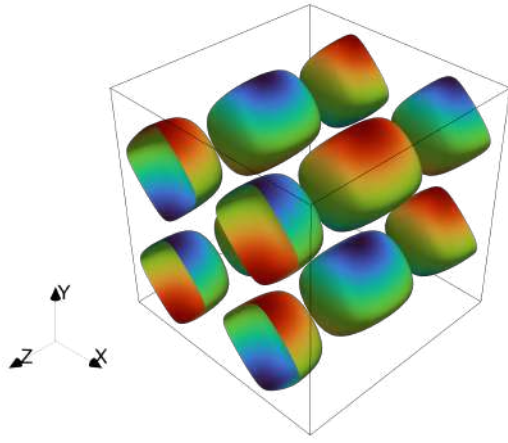
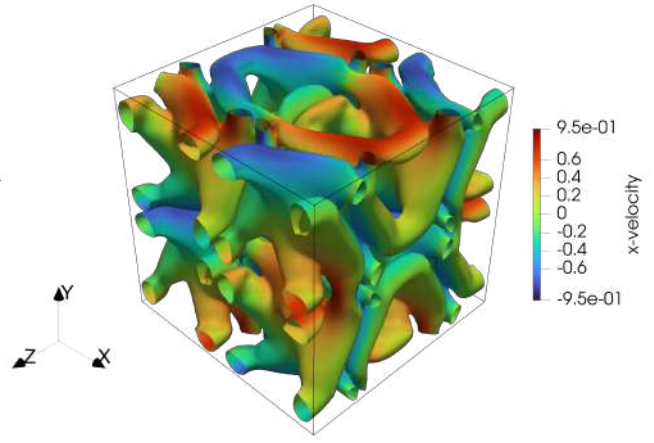


Figure 4: Kinetic Energy Dissipation Rate Variation

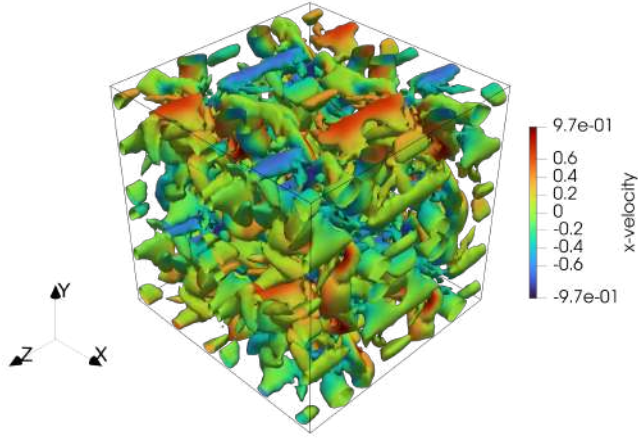
Figures 3 and 4 show large differences in the peak enstrophy and dissipation rate predictions of the 64^3 and 512^3 grid. The deviation of the peak value obtained from the C6F10 set-up from the reference data is around 53.8 %.



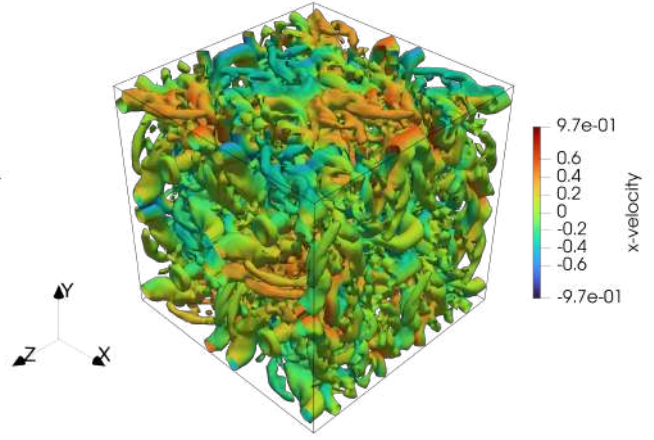
(a) $t^* = 0$



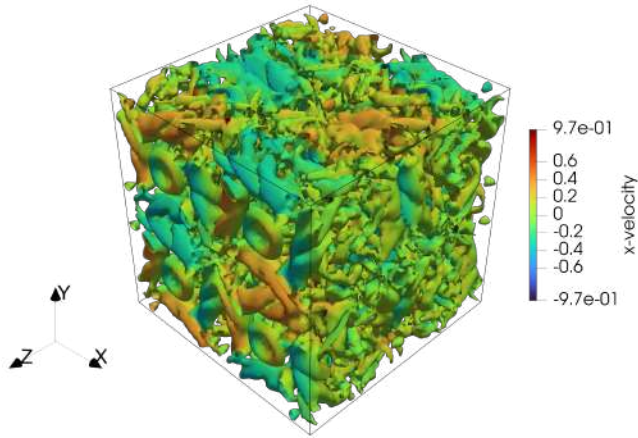
(b) $t^* = 4$



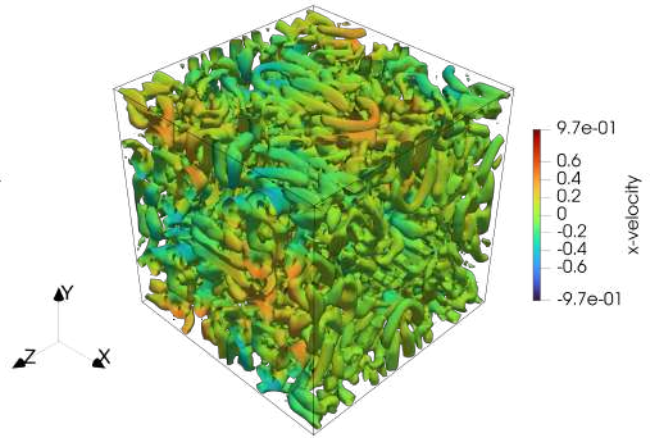
(c) $t^* = 8$



(d) $t^* = 12$



(e) $t^* = 16$



(f) $t^* = 20$

Figure 5: Q-criterion Iso-surfaces

For the C6F10 set-up with $\alpha_f = 0.495$, the isosurfaces of the Q-criterion depicting the magnitude of the i-direction velocity is given in figure 5, where, t^* is the non-dimensional time.

The flow can be seen to be inviscid, having well-defined symmetric vortices. At around $t^* = 4$, the vortices begin to stretch and generate long sheet-like structures. By $t^* = 8$, the structure can be seen to have changed, signifying the onset of turbulence. Subsequent times show the breakdown of coherent structures, leading to the onset of fully turbulent flow.[1]

At $t^* = 8$, on the constant x plane $x/L = -\pi$, the vorticity norm contour plot is shown in figure 6

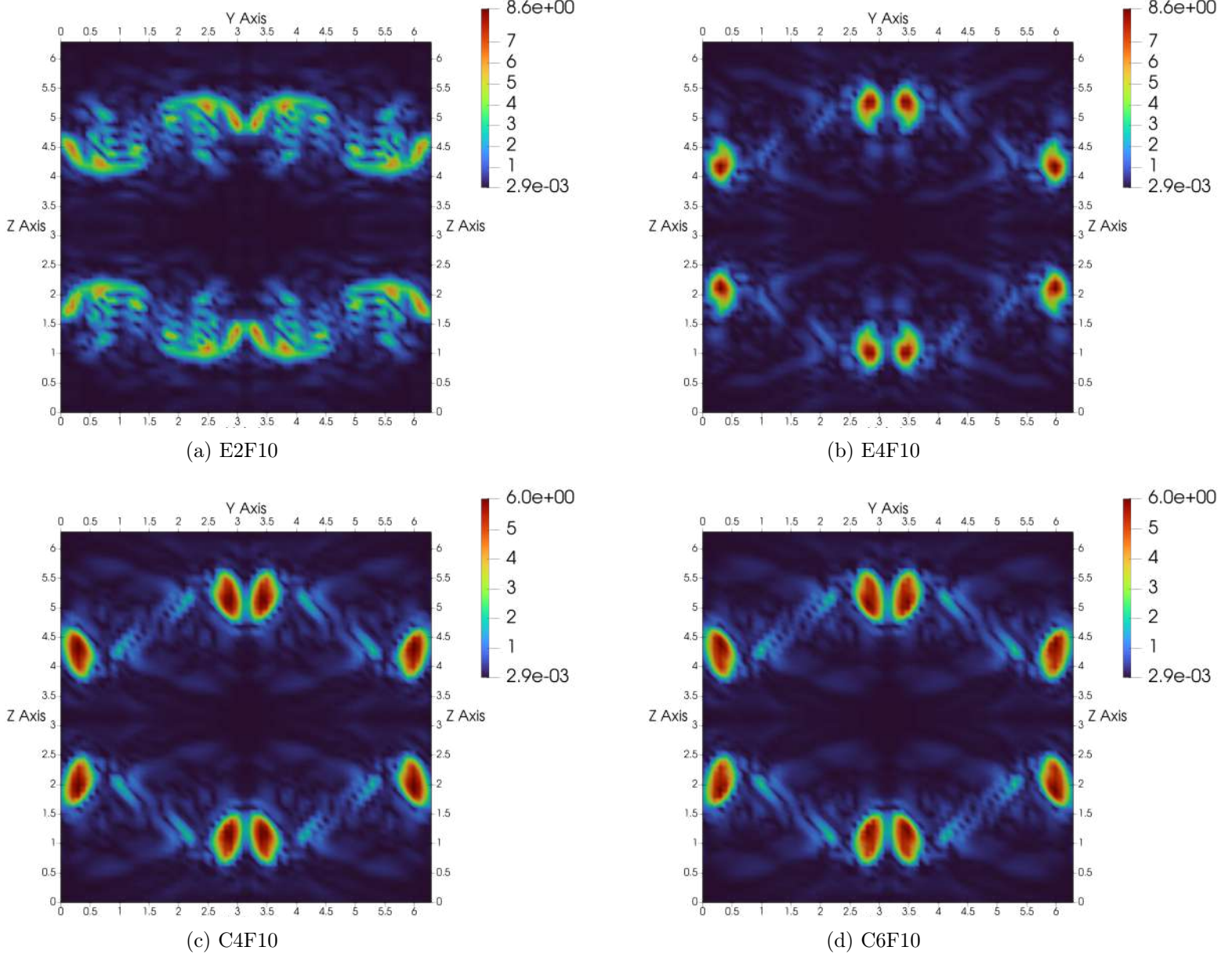


Figure 6: Vorticity Norm Contour Plots

Contours of the vorticity norm show multiple vortices. The maximum non-dimensional vorticity magnitude captured by the compact schemes is 6, while by the explicit schemes is approximately 8.6; E2F10 set-up shows the vortices to be smeared over the large area, and fails to adequately resolve the vortex structures.

The vortex structures captured by the E4F10 set-up is jagged, and not as well-defined as those captured by the compact schemes.

The contours of vorticity on this plane in the region $y/L = 0$ to $\pi/2$ and $z/L = \pi/2$ to π is shown below in figure 7, where a single vortex structure has been isolated for the C6F10 scheme.

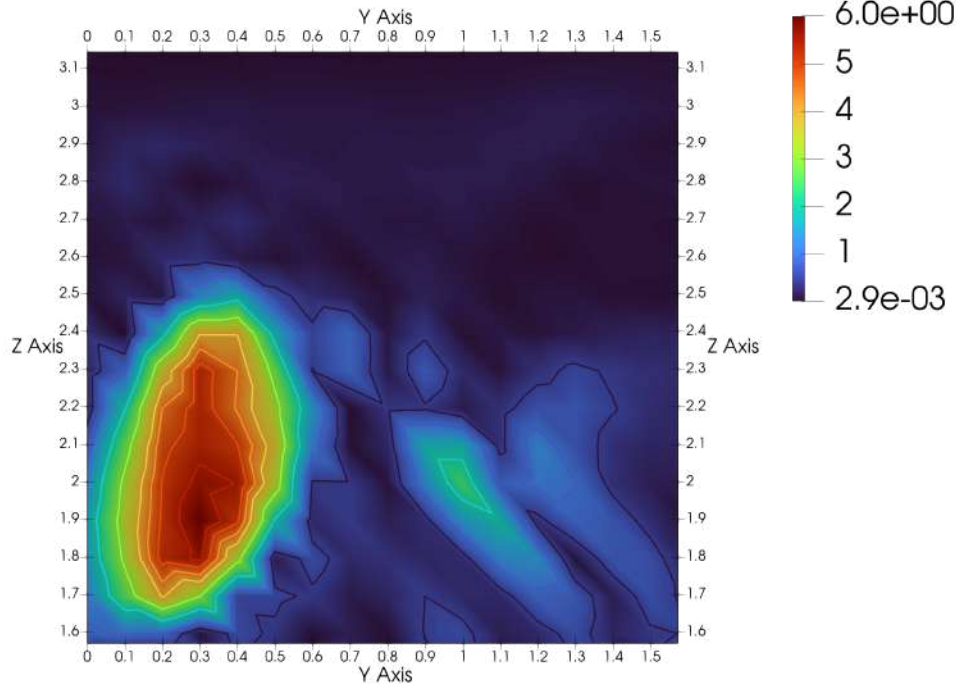


Figure 7: Vorticity norm contour on the plane $x/L = -\pi$ and in the region $y/L = 0$ to $\pi/2$ and $z/L = \pi/2$ to π

2 Inviscid Vortex Convection

2.1 Test Case Summary

The two-dimensional simulation of a vortex convecting in uniform, inviscid, subsonic flow was performed.

Computational Domain	2-D plane with both sides of equal length
Grid	Cartesian, randomly distorted, highly skewed; grid size $\Delta x \in [0.05, 0.4]$
Temporal Discretisation	Runge-Kutta, 4 th order
Time Step	$\Delta t = 0.002$
Space Discretisation	Finite Difference Method, accuracy ranging from 2 nd order explicit to 6 th order compact
Filtering	Implicit filtering, upto 10 th order
Boundary Conditions	Periodic

Table 3: COVO, Test Case Summary

The initial non-dimensional flow conditions are set-up as a vortex centered at the center of the computational domain.

$$u = 1 - Cy \exp\left(-\frac{r^2}{2}\right) \quad (5)$$

$$v = Cx \exp\left(-\frac{r^2}{2}\right) \quad (6)$$

$$p = \frac{1}{\gamma M^2} - \rho C^2 \exp(-r^2) \quad (7)$$

$$r^2 = x^2 + y^2 \quad (8)$$

The non-dimensional vortex strength C is taken as 0.02 and the Mach number of the flow set to 0.1 and the initial density field is taken to be uniform.

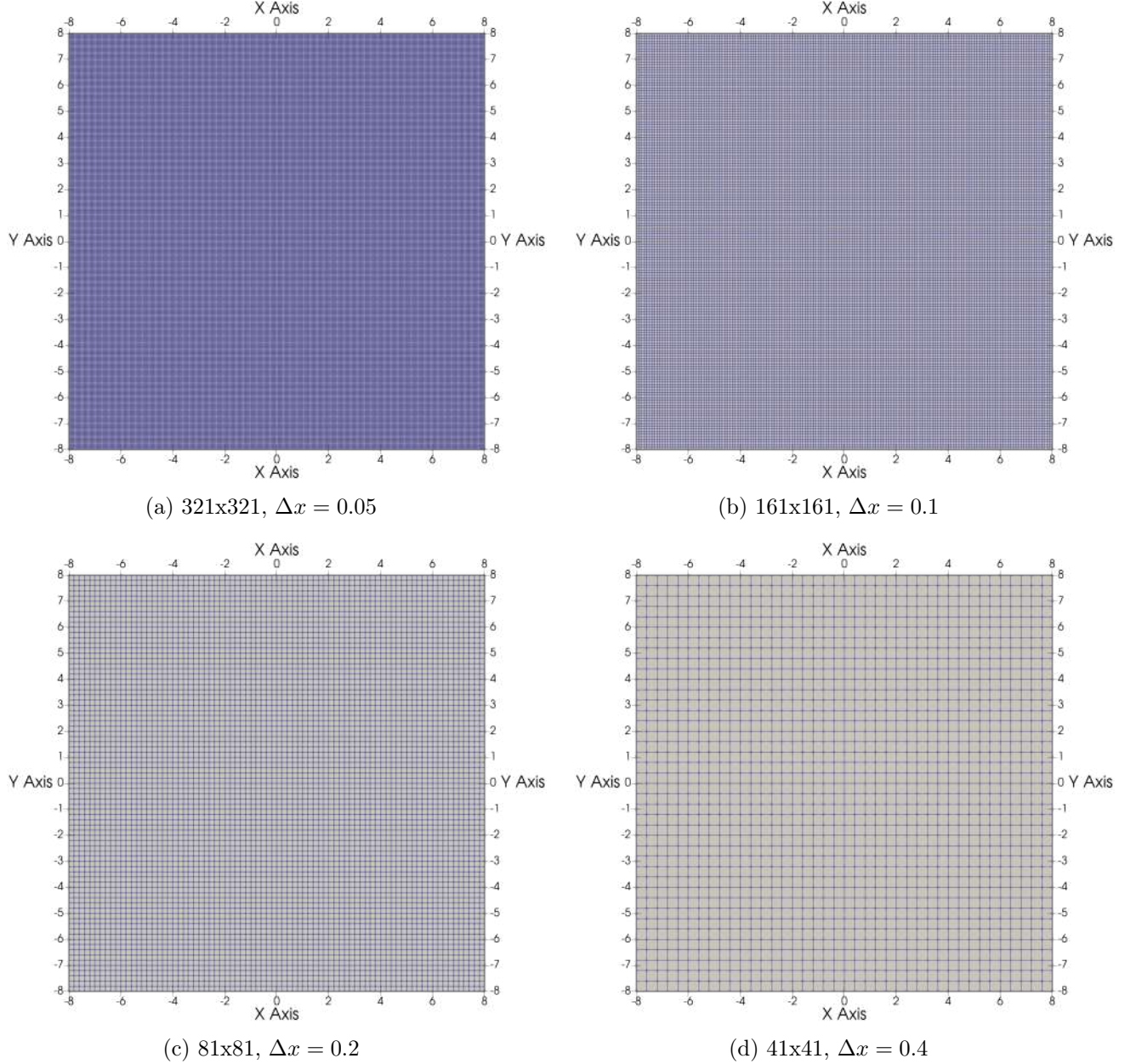


Figure 8: COVO, Computational Domain

2.2 Discussion

Figure 9 shows the contours of the magnitude of vorticity at $t^* = 8$. The vortex strength and structure is seen to be preserved by all the schemes except E2, wherein deformation of the initial vortex can be seen at $t^* = 8$. Figure 10b also shows significant deviation of vorticity from theoretical values for E2.

Calculations were performed with meshes of grid sizes 0.4, 0.2, 0.1 and 0.05 (as shown in figure 8) to determine the maximum error in the computed vertical velocity component along the horizontal line passing through the vortex. The log-log plot of error-vs- Δx and the variation of the magnitude of non-dimensional vorticity is given in figure 10. [2]

All the filtering schemes used $\alpha_f = 0.495$. The error plot contains the obtained error data, and a linear curve fit, the slope is indicated in the legend. The C6 discretization scheme is the most accurate, even for coarser grids (as compared to E2). It was observed that as the grid size decreased significantly, the drop in the error of C6F10 scheme did not go as expected of the 6th order scheme, the slope estimate obtained without considering the finest

mesh was found to be more representative of its accuracy.

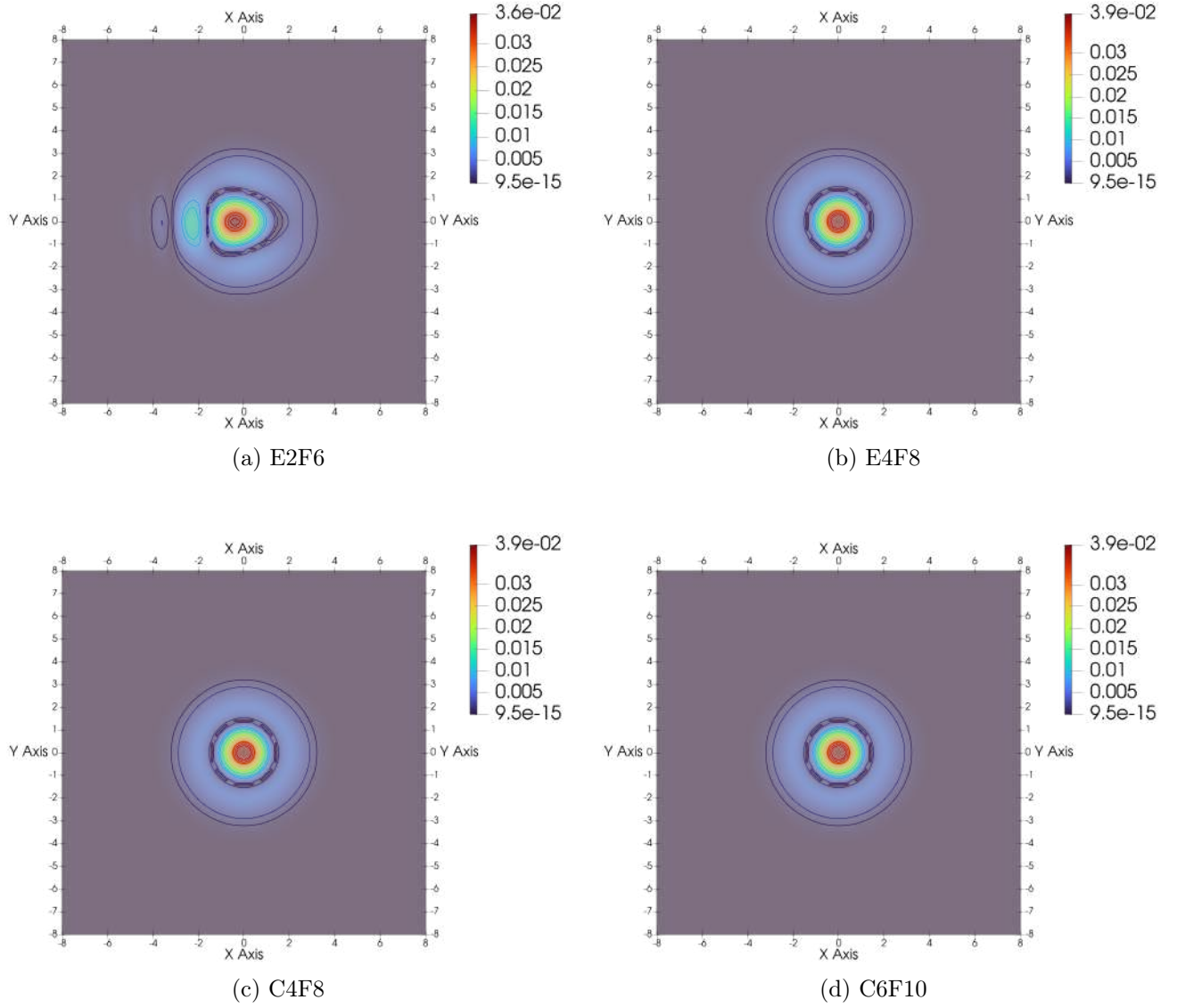


Figure 9: Vorticity Magnitude Contours, $\Delta x = 0.2$, $t^* = 8$

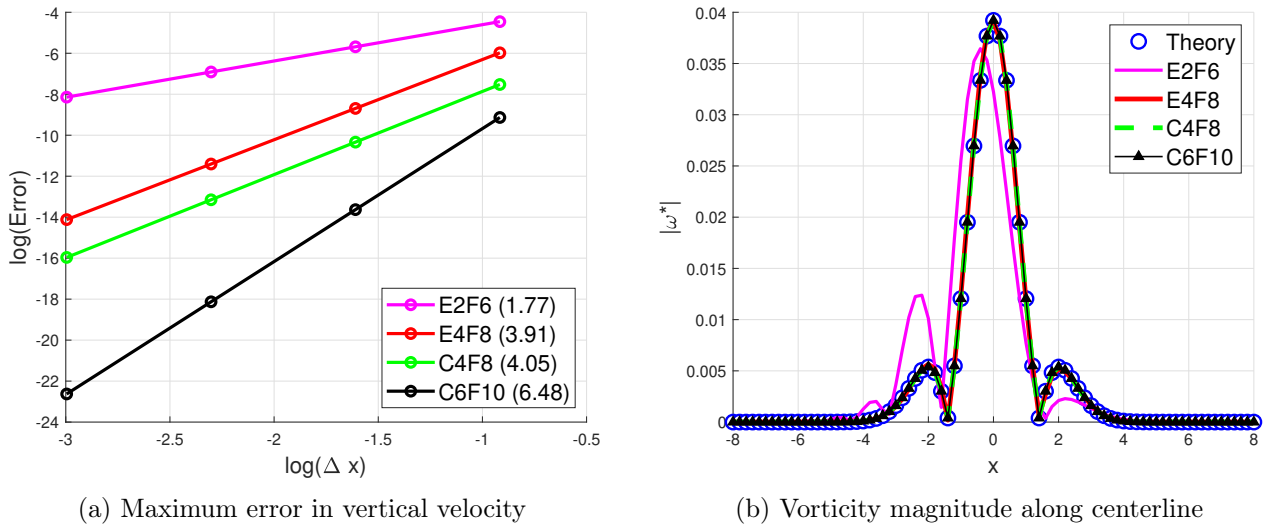


Figure 10: Convection of vortex on uniform mesh with various schemes and grid resolutions at $t^* = 8$

2.2.1 Randomly Distorted Mesh

A randomised 50 x 50 mesh was generated in the domain $-6 \leq x \leq 6$, $-6 \leq y \leq 6$; where the interior points were perturbed by 20% of the grid spacing in a randomly chosen direction. The mesh used is shown in figure 11a and the performance of various schemes on this mesh is given in figure 12. To ensure the stability of the solver, 10th order filters with $\alpha_f = 0.3$ were used. Higher values of α_f weren't able to ensure stability at large time steps. [3]

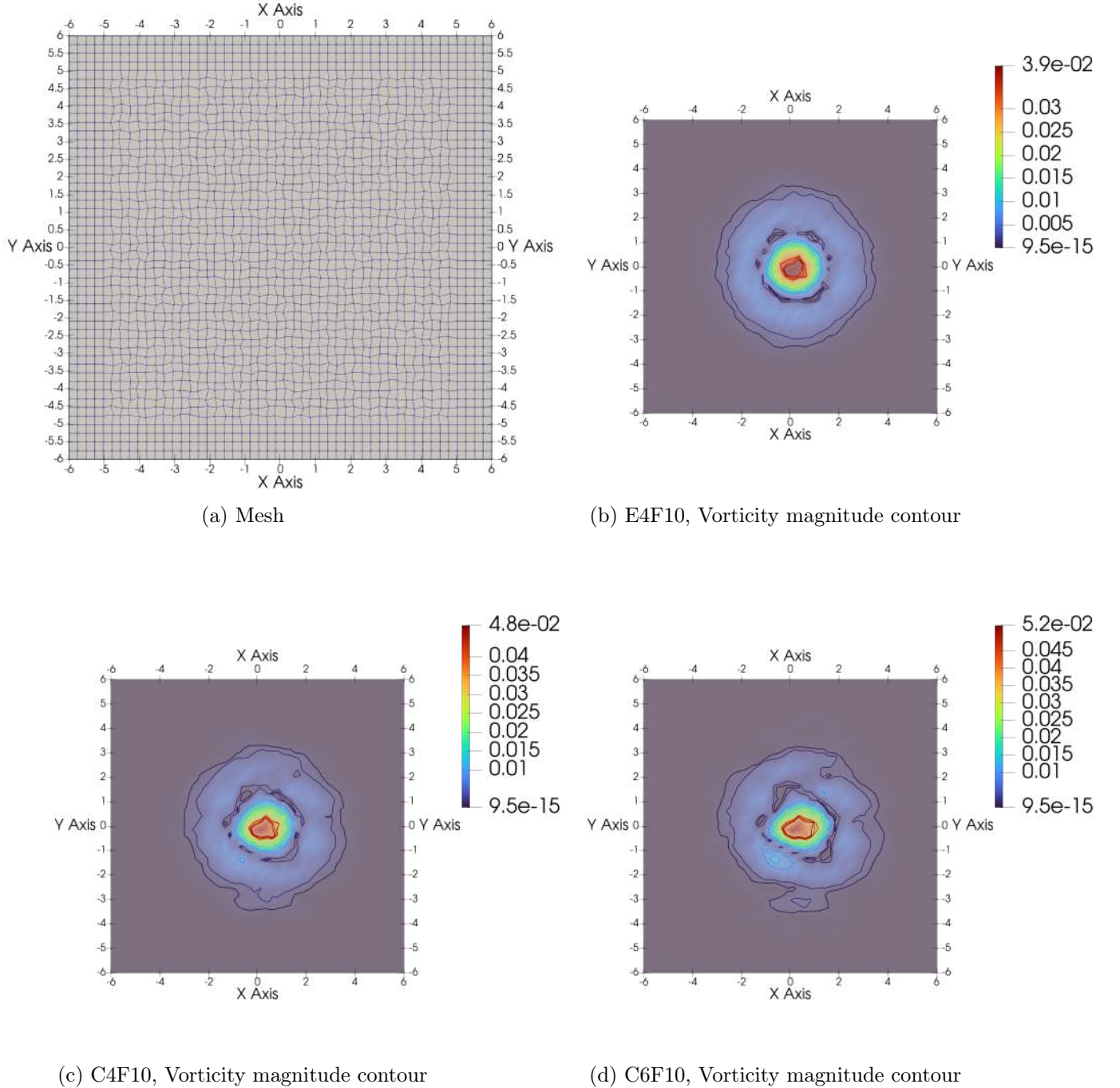


Figure 11: Randomised mesh, $t^* = 6$

The E4 scheme seems to preserve the vortex structure comparatively better than the compact schemes. Significant distortions are seen in the vortex when C6 is used. This observation is reinforced by the plot in figure 12.

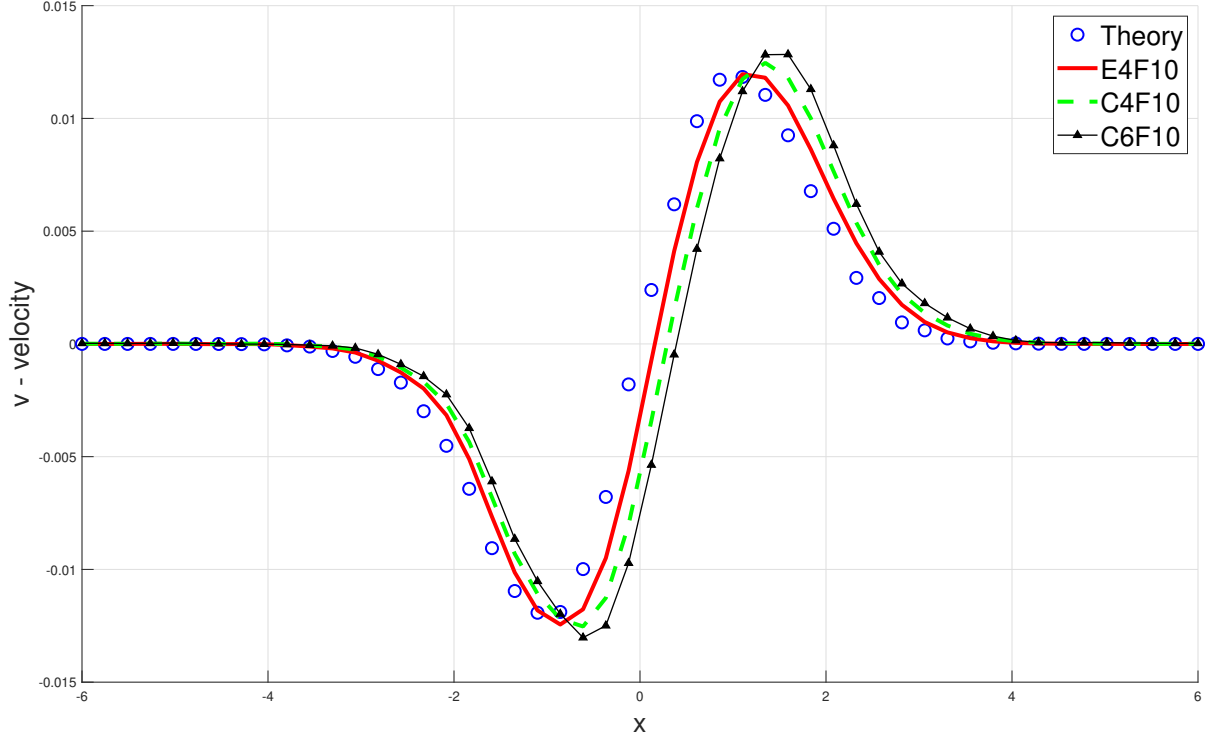


Figure 12: Performance of different schemes on randomised grid

From the figure above a lead error can be seen in the predicted velocity values by the schemes used as opposed to the theory. Moreover, for this randomly distorted grid, the accuracy of compact schemes doesn't seem to be better than the explicit scheme.

2.2.2 Skewed Sinusoidal Mesh

A highly skewed sinusoidal grid was generated with the formula [3]

$$x_{i,j}(\tau) = x_{\min} + x_0(i-1) + A_x \sin(2\pi\omega\tau) \sin\left(\frac{n_x\pi(j-1)y_0}{L_y} + \frac{i\phi_x}{I_L-1}\right) \quad (9)$$

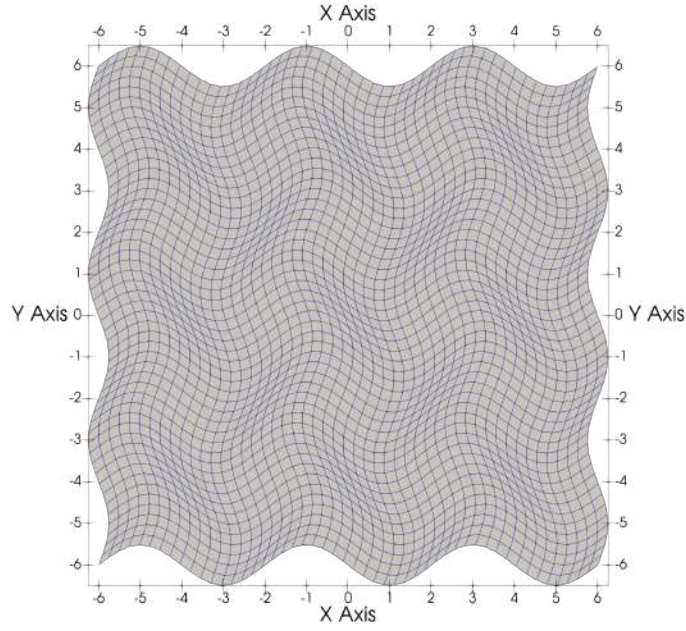
$$y_{i,j}(\tau) = y_{\min} + y_0(j-1) + A_y \sin(2\pi\omega\tau) \sin\left(\frac{n_y\pi(i-1)x_0}{L_x} + \frac{j\phi_y}{J_L-1}\right) \quad (10)$$

$$x_0 = \frac{L_x}{I_L-1}, \quad y_0 = \frac{L_y}{J_L-1} \quad (11)$$

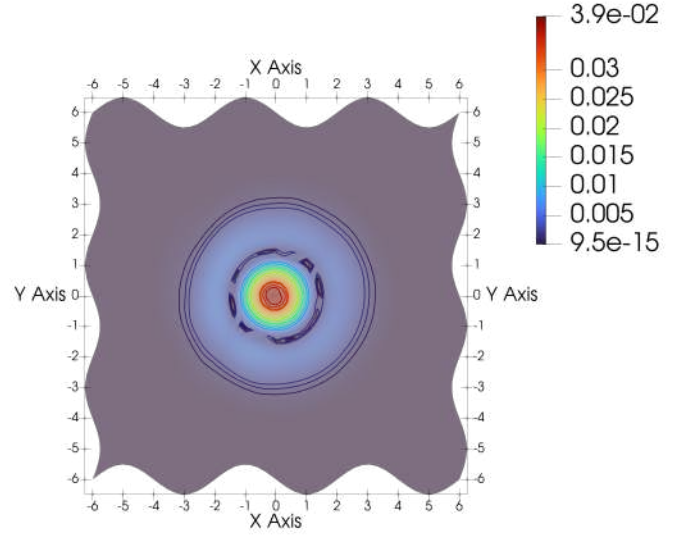
$$1 \leq i \leq I_L, \quad 1 \leq j \leq J_L$$

Where, $x_{\min}$ and $y_{\min}$ are -6, $x_{\max}$ and $y_{\max}$ are 6, L_x and L_y are 12, A_x and A_y are taken to be 1 and 2 respectively, n_x and n_y are taken as 6, ϕ_x and ϕ_y are taken to be 0, $\omega\tau = 1/4$ and I_L and J_L are determined by the number of grid elements. The α_f used for filtering was 0.495. The domain and grid size taken for this case is the same as the distorted mesh case.

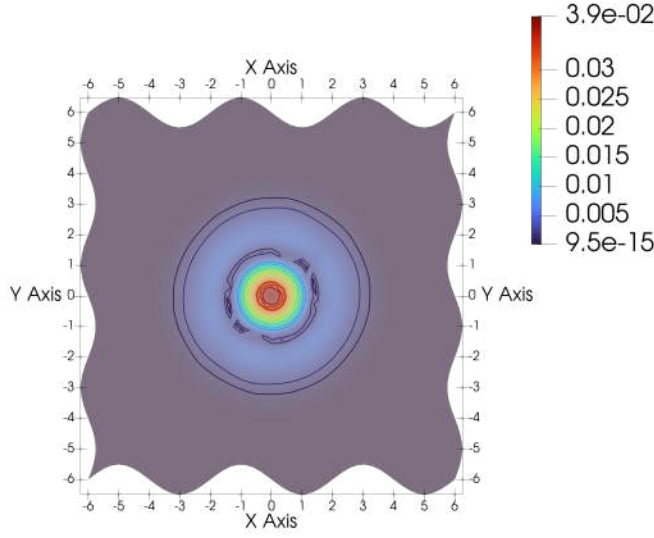
The mesh is shown in figure 13a and the performance of various schemes on this mesh is given in figure 14



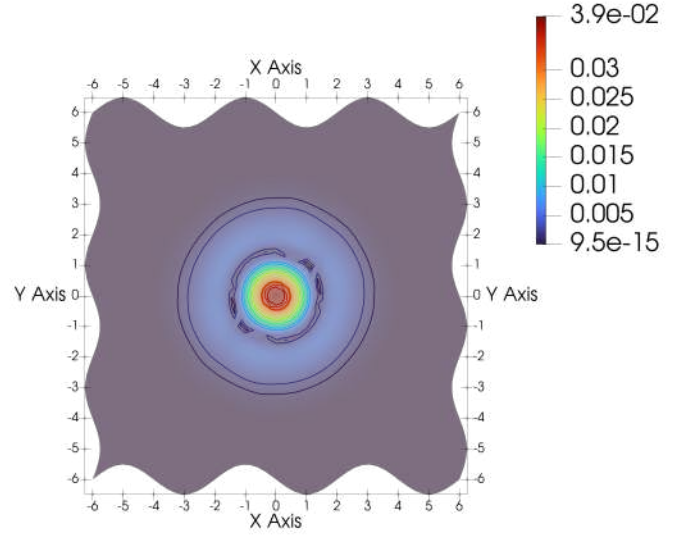
(a) Mesh



(b) E4F10, Vorticity magnitude contour



(c) C4F10, Vorticity magnitude contour



(d) C6F10, Vorticity magnitude contour

Figure 13: Skewed mesh, $t^* = 6$

From figure 13 it can be seen that this mesh helps preserve the vortex much better than the randomly distorted mesh. This could be because the skewed sinusoidal mesh, although not orthonormal has smooth variations as opposed to the random mesh.

From figure 14, it can be seen that at $t^* = 6$, the velocity prediction by the C6 scheme is almost equal to the theoretical prediction. C4 scheme is more accurate than E4.

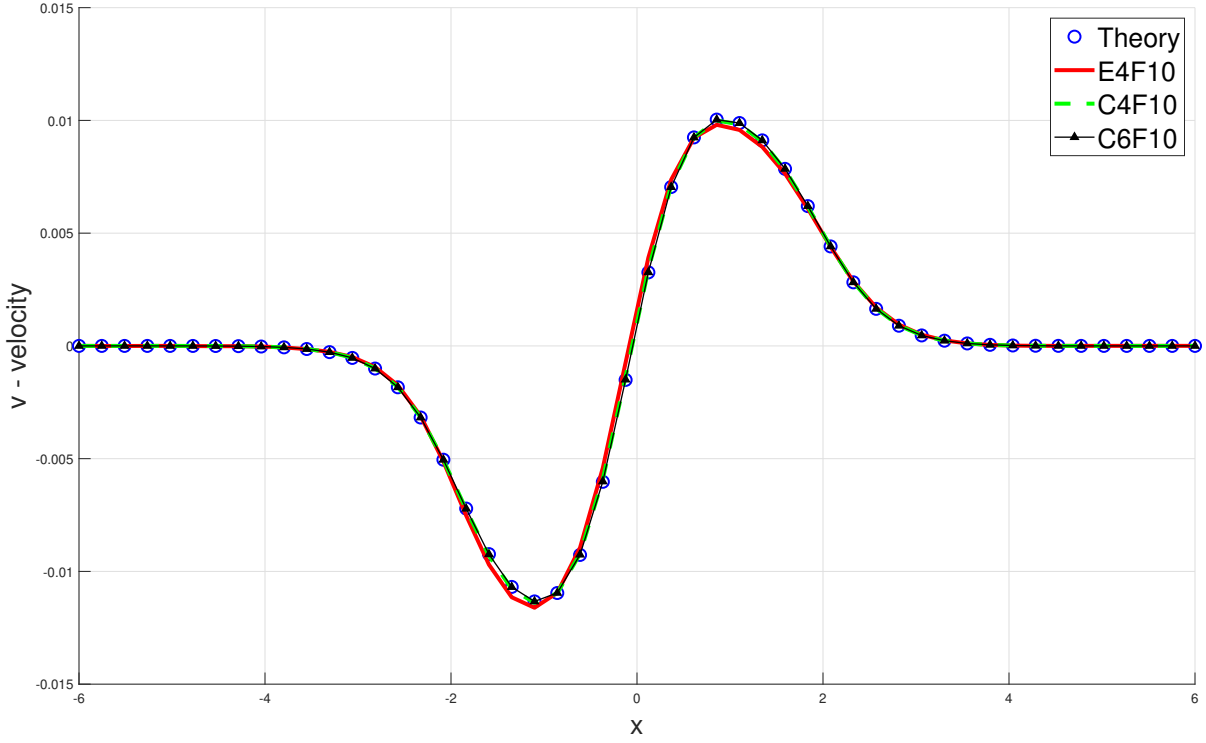


Figure 14: Performance of different schemes on skewed grid

References

- [1] James R. DeBonis. Solutions of the taylor-green vortex problem using high-resolution explicit finite difference methods. In *51st AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition*, Grapevine (Dallas/Ft. Worth Region), Texas, USA, January 2013. American Institute of Aeronautics and Astronautics.
- [2] Miguel R. Visbal and Datta V. Gaitonde. High-order-accurate methods for complex unsteady subsonic flows. *AIAA Journal*, 37(10):1231–1239, October 1999.
- [3] Miguel R. Visbal and Datta V. Gaitonde. On the use of higher-order finite-difference schemes on curvilinear and deforming meshes. *Journal of Computational Physics*, 181(1):155–185, 2002.